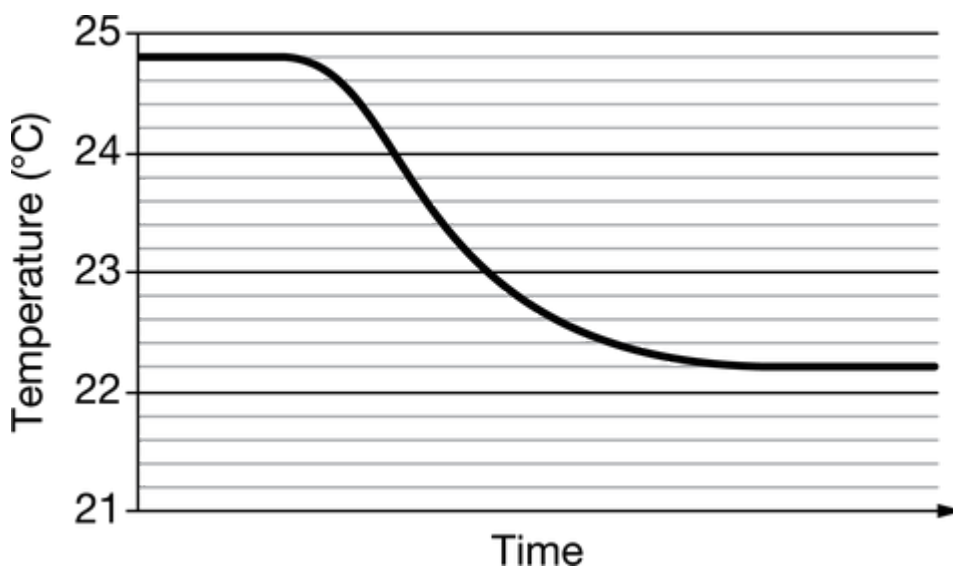


## Unit 6 Progress Check: FRQ

Name \_\_\_\_\_

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student added a sample of  $\text{NH}_4\text{NO}_3(s)$  to water and measured the temperature over time. The data are shown in the graph above.

- (a) What is the value of  $\Delta T$  that the student should use to calculate  $q$ ?



Please respond on separate paper, following directions from your teacher.

- (b) According to the graph, is the dissolution of  $\text{NH}_4\text{NO}_3(s)$  endothermic or exothermic? Explain.



Please respond on separate paper, following directions from your teacher.

- (c) Are the strengths of the interactions between the particles in the solute and between the particles in the solvent before the solute and solvent are combined greater than, less than, or equal to the strengths of the interactions between solute particles and solvent particles after dissolution? Explain.



Please respond on separate paper, following directions from your teacher.

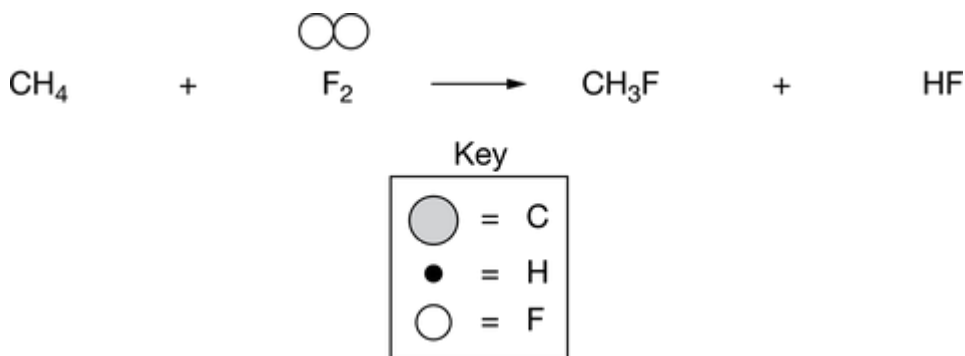


## Unit 6 Progress Check: FRQ

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

$\text{CH}_4(g)$  gas reacts with  $\text{F}_2(g)$  to produce  $\text{CH}_3\text{F}(g)$  and  $\text{HF}(g)$ .

- (a) A particulate representation of  $\text{F}_2(g)$  is shown above its formula in the equation below. Using the key provided, draw particulate representations of  $\text{CH}_4(g)$  and the two product molecules of the reaction above their formulas in the equation.



Please respond on separate paper, following directions from your teacher.

- (b) Use the bond enthalpies in the table below to calculate a numerical estimate of  $\Delta H$  for the reaction.

| Bond  | Bond Enthalpy (kJ/mol) |
|-------|------------------------|
| C – H | 413                    |
| C – F | 485                    |
| H – F | 567                    |
| F – F | 155                    |



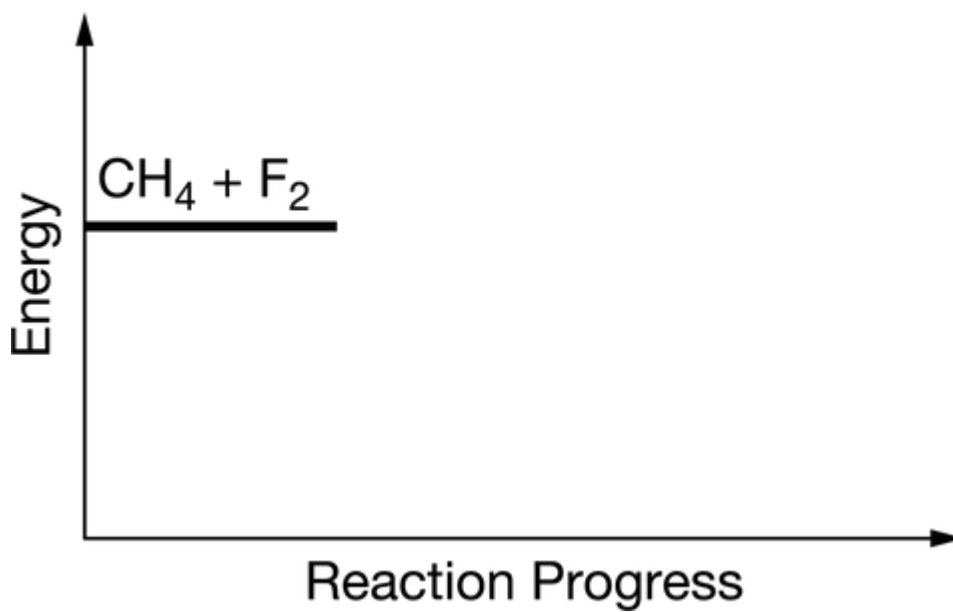
Please respond on separate paper, following directions from your teacher.

- (c) The energy of the reactants is shown on the following energy diagram. On the right side of the energy diagram, draw a horizontal line segment to indicate the energy of the products. Draw a vertical double-headed arrow ( $\updownarrow$ ) on the graph that corresponds to the value of  $\Delta H$  for the reaction.



**Unit 6 Progress Check: FRQ**

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Please respond on separate paper, following directions from your teacher.